MARC ROUSSEAU

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PROFESSIONAL SUMMARY

I'm a Research Engineer with a passion for machine learning applications and data processing systems. During my first year in the field, I've focused on experimenting with neural networks for image classification and building data pipelines that actually work in production. I enjoy diving deep into research papers and translating them into working prototypes, and I'm particularly interested in computer vision problems.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++

Machine Learning: TensorFlow, PyTorch, scikit-learn, OpenCV

Data Engineering: pandas, NumPy, SQL, Apache Spark

Tools and Platforms: Git, Docker, Jupyter Notebooks, Linux

Databases: PostgreSQL, MongoDB

Other: REST APIs, Bash scripting, LaTeX

WORK EXPERIENCE

Research Engineer, Neuroviz.io, Paris, France

January 2024 - Present

Developed a convolutional neural network for medical image segmentation that achieved 94% accuracy on validation dataset while reducing inference time by 35% through model optimization.

Built an automated data pipeline in Python that processes 50,000+ medical images daily, handling format conversion, preprocessing, and quality checks with minimal manual intervention.

Conducted experiments comparing different augmentation strategies for limited training data and documented findings in a technical report that influenced the team's approach to similar problems.

Collaborated with two senior researchers to implement a custom loss function for imbalanced classification, resulting in better performance on minority classes.

Junior Machine Learning Engineer, DataFlow Solutions, Toulouse, France

July 2023 - December 2023

Implemented feature engineering pipelines using pandas and Apache Spark that improved model performance by 12% and reduced computation time from 45 minutes to 8 minutes.

Participated in code reviews and contributed fixes to the production inference service, handling edge cases that occurred with real-world data.

Created evaluation metrics dashboard using Python and PostgreSQL that allowed the team to track model performance across different data subsets in real time.

Assisted in preparing datasets for three different machine learning projects, ensuring data quality and maintaining documentation of preprocessing steps.

Data Science Intern, TechInsight, Lyon, France

February 2023 - June 2023

Explored and visualized customer behavior data using Jupyter Notebooks and matplotlib to identify patterns that informed product recommendations.

Cleaned and prepared 2 million rows of user interaction data, handling missing values and outliers with detailed documentation of decisions made.

Built a simple classification model using scikit-learn to predict customer churn with 81% accuracy on test data.

Presented findings to product team with clear visualizations and actionable insights despite having limited prior experience with stakeholder communication.

EDUCATION

Master of Science in Artificial Intelligence and Data Science

University of Paris-Saclay, Paris, France

Graduation: June 2023

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Statistics, Data Engineering

Bachelor of Science in Computer Science

University of Paris-Saclay, Paris, France

Graduation: June 2021

PUBLIC REPOSITORIES

github.com/marcrousseau/image-segmentation-toolkit

Collection of PyTorch implementations for common image segmentation architectures with training scripts and examples.

github.com/marcrousseau/sentiment-analysis-pipeline

End-to-end data pipeline for processing text data and training transformer models for sentiment classification.

github.com/marcrousseau/neural-network-visualizer

Interactive tool for visualizing how neural networks process images through different layers and activation functions.

github.com/marcrousseau/time-series-forecasting

Experiments with LSTM networks and statistical methods for forecasting time series data from various domains.

LANGUAGES

French: Native

English: Fluent

German: Basic